

Twenty candidate questions, tested

MS 491 Marketing Analytics · Airbnb group assignment · four cities: Antwerp, Amsterdam, Rio de Janeiro, Los Angeles County

This is a shortlist to argue over, not a report. Every question below was run against all four cities before it was written down. Anything that did not survive the test is in the last section with the reason. We need to pick eight: three from the traveller, three from the regional manager, two from the host. Questions are ranked within each perspective, strongest first, so the top three in each block are the default answer unless someone makes a better case.

What "tested" means here. Every number comes from the standard cleaning rules in `findings.py`: drop the 19 listings priced at zero, fill blank `reviews_per_month` with 0 rather than dropping the row (that field is blank exactly when `number_of_reviews` is 0, so dropping would delete 42% of Rio), cap `minimum_nights` at 365, and ignore `last_review` as the brief instructs.

Currency. Prices are EUR in Antwerp and Amsterdam, BRL in Rio, USD in LA. No question compares raw prices. The default normalisation is the **price index** — each listing's price divided by its own city's median — which is unit-free and needs no exchange rate. Where a question needs an actual money comparison, that is stated explicitly and FX and PPP are both reported, because they disagree (question P2.3).

The control that matters. Most comparisons are made *within neighbourhood and room type*, then summarised across cells. This is not fussiness. Pooled comparisons reverse the sign of the professional-host result in Rio and hide it entirely in Amsterdam — a Simpson's paradox we already hit once. Where a result is reported pooled, it is because the within-cell version was checked and agreed.

P1 — The traveller

Someone deciding where to go and what to book. The useful questions here are the ones that change what they actually do: which city, which tier, which neighbourhood, and what they are risking.

P1.1 What does an ordinary budget actually buy — a whole place, or somebody's spare room?

Why someone asks it. This is the first real decision a traveller makes, and it is not about price, it is about privacy. "Can I afford a door I can close?"

Columns. `price`, `room_type`. **Normalisation:** budget expressed as a percentile of the city's own price distribution, so no currency conversion is needed at all.

OF THE STOCK YOU CAN AFFORD, SHARE THAT IS AN ENTIRE HOME	ANTWERP	AMSTERDAM	RIO	LA
Budget at the city's 25th percentile	45.1%	44.1%	40.7%	16.2%
Budget at the median	60.7%	64.4%	57.0%	38.8%
Budget at the 75th percentile	70.6%	72.8%	65.2%	54.1%
<i>Budget percentile needed for a 50/50 shot at an entire home</i>	<i>p34</i>	<i>p28</i>	<i>p37</i>	<i>p66</i>

The cities split two-to-one-to-one. In Amsterdam a below-median budget already puts you mostly among whole apartments; you cross the 50% line at the 28th percentile. In LA you have to spend past the 66th percentile before an entire home is even a coin flip, because private rooms are a third of LA's stock (32.6%) against a fifth of Amsterdam's (20.7%). Antwerp and Rio sit together in the middle. This is the cleanest four-city split we found, and it extends the validated entire-home premium finding by turning a price ratio into something a person can act on.

Trap. The premium itself is not the story — entire homes cost more than private rooms in 100% of the 142 testable neighbourhoods, so that direction is lawlike and boring. The interesting variation is in the *mix* of the market, not the size of the gap. Do not present the ratio; present the ladder.

P1.2 How steep is the ladder? What does it cost to trade up one notch?

Why someone asks it. "If I spend a bit more, do I get meaningfully more, or is everything the same?" It decides whether shopping around is worth the evening.

Columns. `price`. **Normalisation:** price index (price ÷ city median), so every figure is a multiple of local money and directly comparable.

PRICE INDEX AT EACH DECILE	ANTWERP	AMSTERDAM	RIO	LA
10th percentile	0.50	0.54	0.34	0.41
25th percentile	0.71	0.75	0.52	0.64
75th percentile	1.43	1.42	2.15	1.72
90th percentile	2.50	1.94	4.99	3.29
<i>Ratio of the 90th to the 10th percentile</i>	<i>5.0×</i>	<i>3.6×</i>	<i>14.8×</i>	<i>8.0×</i>

Amsterdam is a flat market — the top decile is under 4× the bottom, so trading up buys you very little and the sensible advice is to book the cheap end. Rio is the opposite: the ladder spans nearly 15×, so the same "Rio" holiday means completely different things at different budgets. LA sits in between at 8×. Practically: in Amsterdam, shopping hard saves you little; in Rio it is the whole game.

Trap. Rio's spread is real but partly manufactured by a luxury tail — the maximum is R\$132,358 against a R\$300 median. Use percentiles and medians throughout and never a mean. The p90/p10 ratio is robust to the tail; a standard deviation would not be.

P1.3 What are the odds I am booking a place nobody has ever stayed in — and is that worse at the top of the market?

Why someone asks it. Reviews are the only quality signal in this dataset, and a traveller wants to know their chance of drawing a blank.

Columns. `number_of_reviews`, `price`. **Normalisation:** none needed — the price tiers are within-city quartiles, and the output is a percentage.

SHARE OF LISTINGS WITH ZERO REVIEWS	ANTWERP	AMSTERDAM	RIO	LA
Whole market	18.2%	12.3%	42.0%	21.5%
Cheapest quartile	16.3%	12.7%	25.9%	20.0%
Middle half	13.6%	10.5%	35.4%	18.8%
Dearest quartile	29.6%	15.6%	71.2%	28.3%

The headline is Rio's top quartile: seven out of ten of the most expensive listings in the city have never been reviewed once. That is not a quality signal, it is an indication that Rio's luxury tier is substantially aspirational inventory that has never transacted. Amsterdam is the mirror image — 12.3% overall and barely worse at the top, so almost everything you can book has a track record. The pattern (a U-shape, worst at the expensive end) holds in all four cities but the magnitude ranges from mild to extreme.

Trap. Zero reviews does not prove the listing is bad or fake; a brand-new listing looks identical to an abandoned one in this data. We cannot separate the two without `last_review`, which the brief forbids. Say "no track record", not "fake". This pairs directly with P2.7, which is the same fact read from the platform's side.

P1.4 Can I actually book a weekend here — and does a short stay cost extra?

Why someone asks it. Most city breaks are two or three nights. A listing with a 30-night minimum is invisible to that traveller no matter how good it is.

Columns. `minimum_nights`, `price`, `room_type`, `neighbourhood`. **Normalisation:** the availability figures are percentages; the price comparison is a ratio of medians computed within neighbourhood and room type, so it is unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Bookable for 3 nights or fewer	85.6%	79.3%	70.5%	54.9%
Requires 28 nights or more	2.8%	0.9%	2.0%	32.8%
Price of short-stay vs long-stay stock, within cell	0.93×	1.01×	0.75×	1.15×

Two separate findings, and the second one reverses. First, nearly half of LA's market is closed to a weekend visitor, against one listing in seven in Antwerp — LA's short-term rental ordinance has pushed 30.2% of its listings to exactly 30 or 31 nights. Second, once you compare like with like, short-stay listings in LA cost *15% more* than long-stay ones while in Rio they cost *25% less*. Rio's short-stay stock is the cheap end of the market; LA's is the scarce, regulated, premium end. Amsterdam is flat. That sign reversal is what makes this worth a slide. It survives trimming the top and bottom 1% of prices.

Trap. Antwerp has only six testable cells for the price ratio, so treat the 0.93× as indicative. The availability percentages are solid everywhere.

P1.5 Is the listing everybody books the expensive one?

Why someone asks it. A traveller sorting by popularity wants to know whether that sort is going to cost them money.

Columns. `reviews_per_month`, `price`, within `neighbourhood` × `room_type`. **Normalisation:** a within-cell ratio of medians, so unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Testable cells	13	43	47	147
Price of the busiest quartile vs the rest	0.82×	0.96×	0.68×	0.92×
Cells where the busiest are dearer	15%	35%	0%	30%

No — and in Rio, emphatically not. Rio's busiest quartile is priced 32% below its neighbours in every single one of 47 cells; there is not one exception. Amsterdam is nearly neutral at 0.96× with the busiest being dearer in a third of cells. So the traveller's advice inverts by city: in Rio, popularity and price point the same way and sorting by reviews saves you money; in Amsterdam popularity tells you almost nothing about price. This extends the validated price-versus-reviews finding by turning a correlation into a purchase rule.

Trap. Causation runs both ways and we cannot separate them: cheap listings get booked more, and heavily booked listings may have been priced low to begin with. Present it as an association the traveller can exploit, not as a discount mechanism.

P1.6 Does it pay to shop neighbourhoods, or is the whole city priced the same?

Why someone asks it. "Should I stay one district over to save money?" — and the answer determines whether a traveller should spend any time on the map at all.

Columns. `neighbourhood`, `price`, `room_type`. **Normalisation:** each neighbourhood's median divided by the city's median, entire homes only, neighbourhoods with 50+ listings.

NEIGHBOURHOOD MEDIAN, INDEXED TO THE CITY	ANTWERP	AMSTERDAM	RIO	LA
Neighbourhoods testable	14	20	37	81
10th percentile neighbourhood	0.81	0.83	0.64	0.71
90th percentile neighbourhood	1.19	1.19	2.30	1.49
<i>Dearest neighbourhood ÷ cheapest</i>	<i>1.8×</i>	<i>1.8×</i>	<i>13.9×</i>	<i>10.9×</i>

Antwerp and Amsterdam are geographically flat — moving districts changes your bill by about a fifth either way, so the map is not worth studying. Rio and LA are geographically enormous: the dearest neighbourhood is roughly 11–14× the cheapest. In those two cities, choosing the neighbourhood is choosing the price, and it dominates every other decision the traveller makes.

Trap. Do not explain this with distance from the city centre. We tested that and it fails: the correlation with log price is *positive* in Rio (+0.15, because the value is the beach, not Centro) and flat in LA (+0.08, because LA is polycentric and Malibu is 45km out). The spread is real; the usual explanation for it is not. Name the neighbourhoods instead of drawing a radius.

P1.7 If I need to sleep near the airport, is there anything there — and what does it cost?

Why someone asks it. An early flight, a layover, a one-night business trip. This is a constraint, not a preference, and it is the one case where a traveller ignores the rest of the city.

Columns. `latitude`, `longitude`, `price`, `minimum_nights`, `number_of_reviews`, plus airport coordinates looked up externally. **Normalisation:** price index.

WITHIN 10KM OF THE MAIN INTERNATIONAL AIRPORT	ANTWERP (BRU)	AMSTERDAM (AMS)	RIO (GIG)	LA (LAX)
Listings available	0	8,417 (44.4%)	557 (1.6%)	5,408 (16.4%)
Price index vs city median	—	0.98	0.46	1.18
Bookable for one night	—	15.3%	56.9%	23.2%
Median reviews/month (city figure)	—	0.27 (0.28)	0.00 (0.06)	0.55 (0.44)

Four cities, four completely different answers, which is exactly what we want from a comparative question. Antwerp has *no* listings near Brussels Airport, its actual international gateway 34km away — the question is unanswerable there, and that is itself worth saying out loud about a file labelled "Belgium". Amsterdam is so compact that Schiphol is within 10km of nearly half the city, so "near the airport" is not a distinct segment at all. Rio's airport zone is a genuine budget-transit market: half the city median price, 57% of it bookable for a single night against 28% citywide, but with a median of zero reviews per month, so the supply is there and dead. LA is the only city where the airport zone is a *premium* — 18% above the city median and busier than the city average, and El Segundo, 2.5km from LAX, has the second-highest demand-per-listing of any LA neighbourhood at 2.32×.

Trap, and read this before you pick it. We first tested airport proximity the standard way — as a price gradient across the whole market — and it **fails**. Within neighbourhood and room type the correlation between distance-to-airport and log price is +0.002, +0.021, −0.019 and +0.007 in the four cities: nothing, four times over. It only works as a *conditional* question, asked about the catchment as a segment. If anyone presents this as "listings get cheaper the further you are from the airport", the data does not support it and the slide will not survive a question. Frame it as "who is served near the airport", never as a gradient.

P2 — The Airbnb regional manager

Someone allocating marketing budget across regions and deciding where the platform should push. The useful questions are the ones that identify a problem worth spending money on, and distinguish between "no supply", "supply nobody wants" and "supply that cannot legally be sold".

P2.1 Which city's supply is not converting into bookings — and why not?

Why someone asks it. The core budget question. Adding listings to a market that already cannot sell what it has is wasted money; the fix depends on which kind of failure it is.

Columns. `availability_365`, `reviews_per_month`. **Normalisation:** none — every figure is a share of that city's own listings.

SHARE OF LISTINGS	ANTWERP	AMSTERDAM	RIO	LA
Live — bookable and earning	51.1%	33.4%	44.1%	61.1%
On the shelf — bookable, never booked	11.9%	4.6%	24.5%	15.5%
Blocked — earning, but no calendar open	30.6%	54.3%	13.9%	17.4%
Dead — no calendar, no bookings	6.4%	7.7%	17.4%	6.0%

This is the strongest platform-side question we have, because the four cities land in four different quadrants and each implies a different intervention. Amsterdam's problem is not demand — it is that 54.3% of its listings are earning reviews with a completely closed calendar, which is what a 30-night-per-year cap looks like in data. Advertising into Amsterdam buys nothing; the constraint is regulatory supply, not awareness. Rio is the opposite failure: 24.5% of its stock is sitting bookable and has never sold, plus another 17.4% that is neither bookable nor selling. Rio has too much supply, not too little. LA is the healthiest market at 61.1% live, and Antwerp is middling. The one-line answer to "should I put budget behind Rio instead of Belgium" is no — Rio's problem is demand generation, and Antwerp's is that it is small.

Trap. "Blocked" mixes two different things we cannot separate: a listing sold out and a listing whose host has closed the calendar. Both produce `availability_365 = 0`. In Amsterdam the regulatory reading is well supported because the blocked listings still earn reviews; do not make the same claim for Antwerp without more care.

P2.2 Where inside each city is demand outrunning supply?

Why someone asks it. The actual media-buying decision is made at neighbourhood level, not city level. This finds the neighbourhoods where guests are competing for too few listings — where host recruitment pays — and the ones drowning in unsold stock.

Columns. `neighbourhood`, `reviews_per_month`. **Normalisation:** a ratio of two shares — the neighbourhood's share of the city's monthly reviews divided by its share of the city's listings. 1.00 means supply and demand are matched. Entirely unit-free.

DEMAND SHARE ÷ SUPPLY SHARE, NEIGHBOURHOODS WITH 50+ LISTINGS	ANTWERP	AMSTERDAM	RIO	LA
Neighbourhoods testable	16	22	49	127
Median neighbourhood	0.98	1.00	0.53	0.97
10th / 90th percentile	0.57 / 1.52	0.77 / 1.44	0.25 / 1.25	0.62 / 1.45
Hottest ÷ coldest neighbourhood	2.7×	2.3×	24.2×	9.5×

Antwerp and Amsterdam are internally efficient — every neighbourhood is within roughly a factor of two of balance, so there is no obvious place to point a budget. Rio and LA are wildly unbalanced. Rio's median neighbourhood runs at 0.53, meaning the typical Rio neighbourhood earns half the demand its listing count would justify, while Ipanema (1.55) and Copacabana (1.40) absorb it; Engenho de Dentro runs at 0.06. In LA the spread is 9.5× and the hot spots are not the famous ones — Del Aire (3.01) and El Segundo (2.32) beat Manhattan Beach (0.32) and Bel-Air (0.37) by a factor of seven. That inversion of the obvious answer is the finding. Supply concentration confirms the picture: Rio's largest neighbourhood, Copacabana, is 26.0% of the entire city's listings, against 5.1% for LA's largest.

Trap. `reviews_per_month` is a proxy for bookings, not bookings. It understates demand wherever guests review less often, and that rate may differ by city. The *within-city* comparison is safe; do not compare Rio's 0.53 to LA's 0.97 as if they were the same unit. Compare the spreads, not the levels.

P2.3 "Belgium looks expensive — should I put the budget behind Rio instead?" Which of these markets is actually cheap?

Why someone asks it. Because it is the question that gets asked in the meeting, and because the honest answer is that it depends entirely on a choice most people make without noticing.

Columns. `price`, `room_type`. **Normalisation:** this is the one question that is *about* normalisation, so it reports all three — price index, market FX, and PPP — and shows they disagree.

MEDIAN NIGHTLY PRICE	ANTWERP	AMSTERDAM	RIO	LA
In local currency	€70	€130	R\$300	\$109
Converted at market FX	\$76	\$142	\$61	\$109
Converted at PPP	\$95	\$177	\$124	\$109
Cheapest → dearest at FX	Rio < Antwerp < LA < Amsterdam			
Cheapest → dearest at PPP	Antwerp < LA < Rio < Amsterdam			

Rio moves from cheapest to third-cheapest depending on which conversion you use, and Antwerp and Rio swap ends of the table. Neither conversion is wrong; they answer different questions. FX answers "what does this cost a foreign visitor arriving with dollars", which is the right lens for inbound tourism marketing. PPP answers "what does this cost relative to local incomes", which is the right lens for host recruitment and for judging whether a market is expensive by its own standards. A manager deciding where to advertise to international travellers should use FX and conclude Rio is cheap. A manager deciding where hosts will find listing attractive should use PPP and conclude Rio is not. Amsterdam is the dearest market under every normalisation, which is the only rank that is stable.

Trap. The PPP factors are approximate OECD-style figures and the FX rates are mid-2023, roughly contemporaneous with the July 2023 file dates. The *rank reversal* is robust to reasonable changes in either; the exact dollar figures are not. Present the reversal, and be upfront that the rates are assumptions we brought in from outside the data. This is also the question that justifies why every other question in the deck uses a price index instead.

P2.4 Are private rooms an underused asset or dead weight?

Why someone asks it. Room type is one of the few levers the platform can actually pull through host recruitment and merchandising. If a format is earning more demand than its share of supply, recruit more of it.

Columns. `room_type`, `reviews_per_month`. **Normalisation:** share of demand ÷ share of supply, unit-free. Uses reviews per month, a flow, rather than total reviews, a stock.

	ANTWERP	AMSTERDAM	RIO	LA
Private rooms, share of listings	22.7%	20.7%	25.6%	32.6%
Private rooms, share of monthly reviews	22.4%	47.1%	20.9%	27.5%
Private room index (demand ÷ supply)	0.99	2.28	0.82	0.85
Entire home index	1.02	0.63	1.08	1.11

Amsterdam is a genuine outlier and it reverses the pattern of the other three. Private rooms are a fifth of Amsterdam's listings and nearly half its bookings — they work more than twice as hard as their share of supply, while entire homes work at 0.63. In Antwerp the two formats are in equilibrium, and in Rio and LA entire homes do the heavier lifting. The recommendation writes itself and differs by city: recruit private rooms in Amsterdam, where they are scarce relative to demand and, not coincidentally, are the format least constrained by the entire-home rental cap; do not bother in Rio, where they underperform.

Trap. Amsterdam's number is partly a regulatory artifact, not pure preference — entire-home rentals are capped in a way private rooms are not, so entire homes look weak partly because their calendars are shut. That does not invalidate the recommendation but it does change the reason, and someone will ask. Have the answer ready. It also connects to P2.1: Amsterdam's 54.3% "blocked" share is the same phenomenon seen from another angle.

P2.5 Do professional hosts actually sell better, or do they just charge more?

Why someone asks it. The platform has to decide whether to court property managers or defend the amateur host base. That turns on whether professionals deliver bookings or just extract price.

Columns. `calculated_host_listings_count` (3+ = professional), `reviews_per_month`, `price`, within `neighbourhood` × `room_type`. **Normalisation:** within-cell differences and ratios, unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Professional share of listings	33.4%	9.7%	26.8%	39.2%
Price premium within cell (validated)	+30.0%	+15.7%	−14.7%	+1.1%
Reviews/month gap vs amateurs, within cell	−0.100	+0.040	+0.057	+0.175
Cells where professionals sell more	31%	55%	71%	68%

Putting price and performance side by side is what makes this interesting, and the two rows disagree in a useful way. Antwerp's professionals charge 30% more and sell *less* — they are harvesting a small market, and they are a third of it. LA's professionals charge essentially nothing extra (+1.1%, a genuine null) but sell distinctly more, which looks like operational competence rather than pricing power. Rio's professionals undercut on price by 15% and win on volume. Amsterdam has barely any professionals to speak of at 9.7%. Four cities, four different professional strategies. The platform's answer to "should we court property managers" is yes in LA, no in Antwerp.

Trap. The price row only holds within neighbourhood *and* room type. Pooled, the Rio sign flips positive and Amsterdam's effect disappears behind a room-type composition difference. This is the Simpson's paradox we already documented, and it is worth showing the pooled and controlled numbers side by side rather than hiding it. Also note `calculated_host_listings_count` counts a host's listings on the whole platform, not just in this city.

P2.6 How exposed is each market to a minimum-stay regulation?

Why someone asks it. Regulatory risk is a real line item in a regional plan, and this dataset shows the fingerprint of regulation directly.

Columns. `minimum_nights`, `reviews_per_month`, `availability_365`, `number_of_reviews`. **Normalisation:** shares and within-city medians.

	ANTWERP	AMSTERDAM	RIO	LA
Listings at exactly 30 or 31 nights	1.0%	0.4%	1.2%	30.2%
Median reviews/month, 28+ night listings	0.08	0.14	0.00	0.21
Median reviews/month, 1–3 night listings	0.58	0.33	0.11	0.92
Zero-review share among 28+ night listings	33.8%	26.9%	56.6%	28.4%

LA is the only one of the four with a visible regulatory spike, and it is enormous — 30.2% of listings sit at exactly 30 or 31 nights, a number that cannot occur naturally. That segment earns roughly a quarter of the bookings per month of LA's short-stay stock. So about a third of LA's inventory has been pushed into a format that transacts far less often. The other three cities show smooth minimum-night distributions with no spike at all, which means Amsterdam's supply constraint (P2.1) operates through the calendar rather than through minimum stays — a different regulatory instrument producing a different data signature. For a regional manager this is a map of where the platform's exposure sits and what form it takes.

Trap. We are inferring regulation from a distribution, not reading a statute. The 30-night spike in LA is consistent with its short-term rental ordinance but the dataset cannot prove the causal link. Say "consistent with", and note that the 28+ night segment is a structurally different product (monthly rental) rather than a damaged version of the same one.

P2.7 Is the top of each market real inventory, or a price nobody has ever paid?

Why someone asks it. Reported average prices drive market perception and press coverage. If the expensive end has never transacted, the platform's own headline numbers are describing listings, not sales.

Columns. `number_of_reviews`, `price`, `availability_365`, `calculated_host_listings_count`. **Normalisation:** price index.

LISTINGS THAT HAVE NEVER BEEN REVIEWED	ANTWERP	AMSTERDAM	RIO	LA
Share of the city's stock	18.2%	12.3%	42.0%	21.5%
Their median price index	1.29	1.08	1.66	1.15
Share of them with an open calendar	65.1%	37.6%	58.5%	72.1%
Professional-host share among them (vs city)	44.2% (33.4%)	14.6% (9.7%)	23.1% (26.8%)	40.3% (39.2%)

In every city the never-reviewed listings are priced above the city median, but Rio's are priced 66% above it and they are 42% of the market. Rio's published price distribution is therefore substantially describing inventory that has never sold — which is a serious caveat on any statement about "the price of Rio". Amsterdam is the cleanest at 1.08. Note also who these listings belong to: in Antwerp and Amsterdam they skew professional (44.2% against a 33.4% base; 14.6% against 9.7%), so there the unsold expensive supply is coming from operators, while in Rio it skews amateur. Different problem, different fix — an operator conversation in Antwerp, a pricing-education product in Rio.

Trap. Same limitation as P1.3 — a new listing and an abandoned one are indistinguishable without `last_review`. The 42% figure is a stock of never-transacted listings, not a failure rate. Also worth pairing with P3.2, which is the same fact from the host's side.

P3 — The prospective host

Someone who owns a property and is deciding whether and how to list it. The useful questions produce a number they can act on this week, and an honest statement of how much the data actually pins that number down.

P3.1 "I have a two-bedroom in Amsterdam Oost — what should I charge?" How much do the comparables really tell me?

Why someone asks it. It is the first question every host has. The interesting part is not the median of the comps, which anyone can compute, but how much of the price the comps actually explain — that is, how confident the host is entitled to be.

Columns. `price`, `neighbourhood`, `room_type`. **Normalisation:** R^2 and the residual spread are both unit-free, so they compare directly across currencies.

VARIANCE OF LOG PRICE EXPLAINED	ANTWERP	AMSTERDAM	RIO	LA
Neighbourhood alone	0.122	0.063	0.092	0.277
Neighbourhood $\times$ room type	0.245	0.261	0.240	0.476
<i>Price spread still unexplained, after the comps</i>	<i>1.98$\times$</i>	<i>1.59$\times$</i>	<i>2.59$\times$</i>	<i>1.91$\times$</i>

The honest answer differs sharply by city and it is a genuinely useful thing to tell a host. In LA, knowing the neighbourhood and room type pins down almost half the variation in price, so the comps are trustworthy advice. In Rio they explain less than a quarter, and two otherwise-identical listings in the same neighbourhood and format still differ by a factor of 2.6 — so "look at your neighbours" is weak guidance in Rio and the host is largely on their own. Amsterdam is the tightest market in absolute terms: a low R^2 , but only because there is little variation to explain in the first place, with a residual spread of just 1.59 $\times$. Low R^2 and low spread mean something quite different from low R^2 and high spread, and setting those two numbers next to each other is the contribution here.

Trap. A low R^2 is not a failure of the analysis, it is the finding — it says the observable variables do not determine price, which is exactly what a host needs to know. Do not go looking for more variables to raise it; we do not have the ones that would (size, bedrooms, amenities, photos). Say plainly what is missing.

P3.2 I have no reviews yet. What do hosts like me charge — and are they getting it wrong?

Why someone asks it. Every new host is exactly here, and the instinct is to price at what the property seems worth. The data says what people in that position actually do.

Columns. `number_of_reviews`, `price`, within `neighbourhood` $\times$ `room_type`. **Normalisation:** within-cell percentage difference of medians, unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Testable cells	9	33	79	119
No-review listings priced vs reviewed neighbours	+62.5%	+9.4%	+108.4%	+18.3%
Cells where they are priced higher	100%	67%	97%	83%
<i>Entire homes only, as a robustness check</i>	<i>+43.0%</i>	<i>+8.3%</i>	<i>+173.4%</i>	<i>+22.5%</i>

Hosts with no track record price *above* established neighbours everywhere, which is already counterintuitive — the entrant charges a premium rather than a discount. But the size of the mistake varies by a factor of ten. In Amsterdam the newcomer premium is a mild +9%, and Amsterdam also has the lowest never-reviewed share; in Rio it is +108%, and restricting to entire homes makes it worse at +173%. Read alongside P1.3, this is close to a causal story you can tell a host: Rio's new listings are priced at double the market and 42% of the city's stock has never been booked. The advice is concrete and it is different in each city — in Amsterdam, price at the comps; in Rio, price meaningfully below them until you have reviews.

Trap. The direction of causation is genuinely ambiguous and you should say so. These listings may be unbooked because they are overpriced, or priced high because the host has no feedback yet and is guessing, or they may be a different and genuinely more expensive kind of property that simply arrived recently. The result survives winsorising at 1%/99% and holds within entire homes only, so it is not an outlier artifact — but it is an association, not proof.

P3.3 Should I open my calendar wider? Does being available more of the year actually get me more bookings?

Why someone asks it. Opening the calendar is the cheapest lever a host has — it costs nothing and takes a minute. Hosts assume it works.

Columns. `availability_365`, `reviews_per_month`, within `neighbourhood` × `room_type`, restricted to listings that are bookable and have at least one review. **Normalisation:** correlations, unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Testable cells	9	31	38	126
Median within-cell correlation	+0.028	+0.061	−0.166	−0.085
Cells with a positive correlation	56%	65%	5%	23%

The answer reverses sign across the four cities, which makes it one of the better comparative questions in the set. In Antwerp and Amsterdam more availability is weakly associated with more bookings, as a host would expect. In Rio it is negative in 35 of 38 cells, and in LA negative in 77% of them. The most plausible reading is that the arrow runs the other way: a listing with 300 open days is not aggressively available, it is unsold, while a listing with 40 open days has had the rest of the year booked out from under it. Where that reading holds, high availability is a symptom of failure rather than a cause of success — and a host told "just open your calendar" is being given advice that the Rio data flatly contradicts.

Trap. This is a reverse-causation question and the honest presentation says so explicitly rather than claiming availability harms bookings. The clean version of the claim is: in Rio and LA, availability cannot be used as a lever because it is mostly measuring the outcome, not the input. That is a limitation of the data, and stating it clearly is worth more than pretending the correlation is causal.

P3.4 Is a crowded neighbourhood bad news for me, or is that where the guests are?

Why someone asks it. A prospective host looks at 9,000 listings in Copacabana and wonders whether they are entering a market or a bloodbath.

Columns. `neighbourhood`, `reviews_per_month`, `price`, entire homes only, neighbourhoods with 50+ listings.

Normalisation: correlations across neighbourhoods within a city.

	ANTWERP	AMSTERDAM	RIO	LA
Neighbourhoods tested	14	20	37	81
Correlation of size with bookings per listing	+0.403	+0.404	+0.669	−0.305
Correlation of size with price	+0.258	+0.497	−0.137	+0.180

Three cities say competition is a good sign and one says the opposite. In Antwerp, Amsterdam and especially Rio, listings in bigger neighbourhoods get *more* bookings each, not fewer — supply has gone where the guests are and the crowd is a signal to follow. LA alone is negative: its big neighbourhoods are big because LA County is enormous and residential, not because they are tourist destinations, so scale there dilutes rather than concentrates. Amsterdam adds a second twist — its busy neighbourhoods are also its expensive ones (+0.497), so there you pay for the crowd, while in Rio the busy neighbourhoods are slightly *cheaper* (−0.137) and a host gets volume without paying for location.

Trap. This is neighbourhood-level, so *n* is small — 14 neighbourhoods in Antwerp, and correlations on 14 points are fragile. Quote Rio and LA with confidence and Antwerp with care. It is also the constructive counterpart to our double jeopardy test, which failed: treating neighbourhoods as brands does not reproduce the Ehrenberg pattern, and we report that failure deliberately.

P3.5 Does it matter whether I price at a round number?

Why someone asks it. Price endings are a classic marketing question and hosts pick them by instinct. Nobody else in the class will look at this column this way.

Columns. `price`, `reviews_per_month`, `calculated_host_listings_count`, within `neighbourhood` × `room_type`.

Normalisation: shares and within-cell differences. The digit itself is currency-dependent, which is part of the finding.

	ANTWERP	AMSTERDAM	RIO	LA
Price ends in 0	38.5%	43.5%	12.5%	30.0%
Price is a multiple of 50	13.1%	18.9%	7.4%	13.5%
Reviews/month gap, round vs not, within cell	−0.228	−0.110	+0.020	−0.325
Cells where round-priced listings do better	25%	13%	65%	8%

Two things vary and they vary independently. First, how common round pricing is: 43.5% of Amsterdam listings end in a zero against 12.5% in Rio, which is largely a denomination effect — a round number in euros is a coarser instrument than a round number in reais, where the median price is 300. Second, and more interesting, whether it is associated with anything. In LA round-priced listings are meaningfully less booked in 92% of cells; in Rio they are slightly *more* booked in 65% of cells. So the same superficial habit carries opposite associations in the two largest markets.

Trap. The obvious confound is that round pricing is a proxy for an inattentive host, and inattentive hosts do worse for many reasons that have nothing to do with the digit. We checked whether it is simply professionals versus amateurs and it is not a clean split — in LA round-priced listings are 30.4% professional against a 39.2% base, but in Rio the shares are nearly identical (27.2% against 26.8%). So the confound is present but does not explain the reversal. Rank this last of the P3 questions: it is the most novel and the least certain, and it should only make the final eight if the team wants one genuinely unexpected slide.

P3.6 Does a longer minimum stay cost me bookings?

Why someone asks it. Minimum nights is a setting a host chooses on day one, usually to reduce turnover effort, and they rarely revisit it.

Columns. `minimum_nights`, `reviews_per_month`, within `neighbourhood` × `room_type`. **Normalisation:** correlations and within-city medians, unit-free.

	ANTWERP	AMSTERDAM	RIO	LA
Median within-cell correlation	−0.229	−0.144	−0.131	−0.314
Cells with a negative correlation	92%	93%	96%	96%
Median reviews/month at 1 night	0.61	0.33	0.02	0.86
Median reviews/month at 4–7 nights	0.19	0.18	0.02	0.27
Median reviews/month at 28+ nights	0.08	0.14	0.00	0.21

Yes, everywhere, and that is both the strength and the weakness of this question. The direction is negative in 92–96% of cells in all four cities, so the advice is universal: raising your minimum stay costs you bookings. What differs is how much. In LA the penalty is steepest (−0.314) and the drop from a one-night to a four-to-seven-night minimum is roughly two-thirds of your booking rate. In Rio the correlation is weakest and the levels are so low throughout that the lever barely matters. So the useful comparative framing is not "does it cost you" but "how much of your booking rate is this setting worth", which ranges from about a third to about two-thirds.

Trap, and the reason this ranks last. The answer is qualitatively the same in all four cities, which by our own criterion makes it a weaker comparative question than the five above it. Include it only if the team wants a question that gives a host a clear, confident, immediately actionable instruction — which it does, and which none of the others quite do. Note also that the causal direction is defensible here in a way it is not for P3.3: a host sets minimum nights before guests respond to it.

What we tested and killed

Everything here was actually run against all four cities. We are listing it because a failed test is harder to fabricate than a confirmed one, and because two of these are the analyses a group would reach for first.

Distance from the city centre. The standard Airbnb analysis, and it is simply wrong on this data. The correlation with log price is *positive* in Rio (+0.15 — the value is the beach, not Centro) and flat in LA (+0.08 — LA is polycentric and Malibu is 45km out). Antwerp has exactly one listing beyond 10km, so there is no gradient to measure. Do not let this back into the deck.

Distance to the nearest major airport, as a price gradient. The idea we were asked to evaluate. Tested against ANR and Brussels for Antwerp, Schiphol, Galeão and Santos Dumont, and LAX, Burbank, Long Beach and John Wayne. Pooled, the correlations with log price are +0.117, -0.019, +0.167 and -0.045 — weak and inconsistent in sign. Within neighbourhood and room type they collapse to +0.002, +0.021, -0.019 and +0.007: nothing at all, in all four cities. There is also no variation to exploit in the European cities — 80% of Antwerp's listings sit between 3.0 and 5.6km from its airport, and Amsterdam's between 8.0 and 13.3km. **The gradient version is dead.** What survives is the conditional version, P1.7 — the airport catchment treated as a segment with its own price level, stay length and demand — and that one is genuinely strong because the four cities differ completely. Do not confuse the two; the distinction is the whole finding.

Distance to the supply centroid, our attempt to rescue a geography variable by defining "the centre" from where the listings actually are rather than from a map. Within neighbourhood and room type the correlations are -0.096, -0.052, -0.032 and +0.011 — weak, and null in LA. Not worth a slide. Geography in this dataset works through named neighbourhoods (P1.6, P2.2), not through radius.

Total review count as a demand measure. It mostly measures how long a listing has existed. Dividing reviews by reviews-per-month gives an implied months-live that ranges from about 5 to about 65 months within every city, and log reviews correlates +0.44 to +0.54 with that implied age. Use `reviews_per_month`, a flow, wherever demand is meant.

Estimated revenue per listing. Requires an assumed review rate and an assumed stay length, neither of which is in the data. Any number produced this way is an assumption wearing a result's clothing.

Top-10 host share. A market-size artifact — it falls automatically as a market gets bigger. Share of listings from hosts with 3+ listings is the honest version and is used in P2.5.

Shared rooms and hotel rooms as segments. Too small to compare. Shared rooms are 0.3% of Amsterdam and 3.5% of LA; hotel rooms are 0.4% to 1.2% everywhere. Antwerp has 33 shared and 15 hotel listings in total, and zero neighbourhoods with 50 or more of either. Any four-city comparison would be reading noise in two of the four cities.

Neighbourhood level 1 (`neighbourhood_group`). Empty in three of the four files. It has three values in LA and nothing anywhere else, so it cannot support a comparative question.

Listing title in English versus the local language. A genuinely interesting positioning question that we wanted to work, and it half does. Rio is the only city with a real split at 42.7% English titles, and there the effect on bookings is exactly nothing — a rpm gap of +0.000, higher in 46% of 76 cells. Amsterdam (+0.050) and LA (+0.130) lean positive but are 94% and 86% English already. Antwerp is untestable: 85.6% English and only four mixed cells. A question we can only answer in three cities, with a null in the one city where it matters, is not worth one of eight slots — but the Rio null is a nice throwaway line if someone asks about positioning.

Keyword premia in listing titles. Tested 17 keywords within neighbourhood and room type. Only one signal is consistent across cities: "studio" in the title is a discount everywhere ($0.52\times$ to $0.79\times$), which is unsurprising since it describes the property, not the marketing. Everything else is thin — "view" ranges from $1.03\times$ in Amsterdam to $1.41\times$ in LA on 14 to 34 cells, and "beach" in Rio comes out at $0.90\times$, below neutral. Too few observations per cell to defend.

Double jeopardy, treating neighbourhoods as brands. Retained deliberately as a reported failure rather than dropped. The correlation between neighbourhood size and median reviews per month is +0.44, -0.09, +0.65 and -0.21 — no consistent pattern, so the law does not hold here. It is worth one slide precisely because it did not work, and P3.4 is the constructive question built on the wreckage.

The dashboard

Recommendation: build it for the host (P3), driven by P3.1, P3.2 and P3.6.

The test is which perspective is naturally *parametric* — where changing an input produces a genuinely different answer rather than a filtered subset of the same one. Only the host passes cleanly. A host has a specific property in a specific neighbourhood at a specific format, and every one of those choices moves the answer. The regional manager's questions are a fixed comparison of four markets; there is one right answer per city and it belongs on a slide, not behind a control. The traveller's questions are parametric too, but a traveller dashboard collapses into exactly the filter-and-histogram we should avoid: set a budget, see the listings that match. That is a search box, and Airbnb already has one.

The host case is different because the output is not a subset of listings, it is a *decision with a stated confidence*. That is the thing a static slide cannot do, because the confidence depends on inputs we cannot enumerate in advance.

Inputs. City; neighbourhood; room type; intended minimum nights; and a candidate nightly price the host types in.

Outputs. Four, in this order:

- **Where that price sits** in the distribution of genuine comparables — same neighbourhood, same room type — as a percentile, with the median and quartiles shown. The obvious part, and the part every dashboard does.
- **How much that comparison is worth.** The residual spread from P3.1: "in this neighbourhood and format, otherwise-identical listings still differ by 2.6×, so treat the median as a starting point, not an answer." This number changes with the inputs and it is the reason the dashboard has to be interactive. It is also the output no static chart can carry, because it is a statement about the strength of its own advice.
- **The minimum-nights penalty,** from P3.6: the median reviews per month for that cell at the chosen minimum stay against 1 night, so the host sees what the setting costs before they choose it.
- **The newcomer warning,** from P3.2: if the host has no reviews, show what other no-review listings in that cell charge and what fraction of them have never been booked. In Rio this output is dramatic and directly actionable; in Amsterdam it is nearly silent. That the same control produces a loud answer in one city and a quiet one in another is the argument for building it.

What the user actually learns. Not "here is the price" — they learn how much the market determines their price and how much is left to them, which is different in every city and every neighbourhood, and which is invisible in any single view. A host in LA leaves knowing the comps are trustworthy. A host in Rio leaves knowing the comps are nearly useless, that the typical newcomer there is priced at double the market, and that 42% of the city's listings have never been booked. Those are opposite pieces of advice from one tool, which is the case for the tool.

Argument against the generic version. A filter-and-histogram dashboard answers "what is in the data". This one answers "what should I do, and how much should I trust it". The distinguishing feature is that it reports its own uncertainty as a first-class output rather than hiding it, which is also the learning outcome the course is assessing. If we build a price slider and a bar chart, we have built something fifteen other groups could build from the same file.

Picking the eight

Default recommendation, if nobody objects: **P1.1, P1.3, P1.4** for the traveller — the budget ladder, the odds of a listing with no track record, and weekend bookability with its price reversal. **P2.1, P2.2, P2.3** for the manager — the inventory-health quadrants, the neighbourhood demand-supply mismatch, and the currency question that answers "should I back Rio instead" with a rank reversal. **P3.1, P3.2** for the host — what the comps are worth, and the newcomer overpricing result.

That set covers all four cities in every question, contains three sign reversals and one rank reversal, extends the validated findings in four of the eight, and leaves P1.7 and P3.5 as the two most distinctive substitutes if the team wants something less conventional. The two arguments most likely to be made against it: swapping P1.7 in for P1.4 to get the airport question in front of the class, and swapping P2.5 in for P2.3 to lead with the Simpson's paradox instead of the currency reversal. Both are defensible.